

Exact Bloch point evaluation on the unit polydisk

Candidate full solution packet for arXiv:2208.02147

9 August 2026

Status: candidate full solution, likely valid pending expert review.

Abstract

Allen asked for a closed form of the norm $\omega(z)$ of point evaluation at z on the normalized Bloch space of the unit polydisk. We prove

$$\omega_0(z) = \omega(z) = \rho(z, 0) = \left(\sum_{j=1}^n \operatorname{arctanh}^2 |z_j| \right)^{1/2}.$$

Thus the previously known Poincare-distance upper bound is exact. The lower bound is attained in the Bloch space by an explicit Euclidean-normalized sum of rotated one-variable hyperbolic arctangents. Radial dilations of the same function belong to the star-little Bloch space and converge to the extremal value, giving the stronger ω_0 identity. Substitution yields an exact symbol formula for the norm of the weighted composition operators studied in the source.

1 Source question

The source is R. F. Allen, *Weighted composition operators from the Bloch space to weighted Banach spaces on bounded homogeneous domains*, arXiv:2208.02147 [1]. Open Question 1 on PDF page 9 asks whether $\omega(z)$ has a closed form on $\mathbb{D}^n$, and whether there is a sharper upper bound than the Poincare distance $\rho(z, 0)$.

6. CONCLUSIONS AND OPEN QUESTIONS

The general bounded homogeneous domains offer a framework by which traditional operator theory on spaces of holomorphic functions in several complex variables (specifically on the unit ball and the unit polydisk) can be unified. As with any generalization, any results we gain are met with trade-offs. One of the trade-offs is the use of the $\omega(z)$ in characterizing boundedness and compactness of the multiplication, composition, and weighted composition operators. For the unit ball, a closed form for $\omega(z)$ is known, and upper bounds are known for the unit polydisk. However, no closed form for the polydisk, or no bounds known for other bounded homogeneous domains presents a temporary roadblock. This roadblock presents itself when we consider conditions for compactness of the composition and weighted composition operators.

We conclude this paper with the following open questions we hope researchers will take up.

- (1) Is there a closed form for $\omega(z)$ on $\mathbb{D}^n$? Is there a sharper bound than $\rho(z, 0)$?
- (2) Are there other specific bounded homogeneous domains for which a closed form for $\omega(z)$ can be determined?
- (3) Is Conjecture 4.2 true for all bounded homogeneous domains? Is Conjecture 4.2 true for any other specific bounded homogeneous domain?

REFERENCES

Figure 1: The source's open questions on page 9 of arXiv:2208.02147.

For $w \in \mathbb{D}^n$ and $f \in H(\mathbb{D}^n)$, the source uses the metric

$$H_w(u, \bar{u}) = \sum_{j=1}^n \frac{|u_j|^2}{(1 - |w_j|^2)^2}$$

and the Bloch seminorm $\beta_f = \sup_w Q_f(w)$, where $Q_f(w)$ is the dual norm of df_w . Its normalized evaluation quantities are

$$\begin{aligned} \omega(z) &= \sup\{|f(z)| : f \in \mathcal{B}(\mathbb{D}^n), f(0) = 0, \beta_f \leq 1\}, \\ \omega_0(z) &= \sup\{|f(z)| : f \in \mathcal{B}_{0*}(\mathbb{D}^n), f(0) = 0, \beta_f \leq 1\}. \end{aligned}$$

The source proves

$$\omega_0(z) \leq \omega(z) \leq \rho(z, 0). \quad (1)$$

2 Proof intuition

The product metric dictates an ℓ_2 answer. The hyperbolic distance in the j th disk is $t_j = \operatorname{arctanh} |z_j|$, so the distance in the product is $R = (\sum_j t_j^2)^{1/2}$. In one variable, $\operatorname{arctanh}(\eta w)$ has Bloch seminorm one and takes the value t_j at z_j when $\eta z_j = |z_j|$. Weight these one-variable extremals by the Euclidean unit vector $(t_1/R, \dots, t_n/R)$. The dual metric makes the resulting Bloch seminorm at most one, while its value at z is exactly R .

For the star-little Bloch space, replace $\operatorname{arctanh}(\eta w)$ by $\operatorname{arctanh}(s\eta w)$ with $s < 1$. These dilates extend past the closed disk, remain in the same Bloch unit ball, and their target values tend to the same R as $s \uparrow 1$.

3 The metric and its extremal

Lemma 1. For $z = (z_1, \dots, z_n) \in \mathbb{D}^n$,

$$\rho(z, 0) = R(z) := \left(\sum_{j=1}^n \operatorname{arctanh}^2 |z_j| \right)^{1/2}.$$

Proof. Let $\gamma : [0, 1] \rightarrow \mathbb{D}^n$ be any piecewise C^1 path from 0 to z , and put

$$v_j(t) = \frac{|\gamma'_j(t)|}{1 - |\gamma_j(t)|^2}.$$

Minkowski's integral inequality and the one-variable disk distance give

$$\operatorname{length}(\gamma) = \int_0^1 \left(\sum_j v_j(t)^2 \right)^{1/2} dt \geq \left(\sum_j \left(\int_0^1 v_j(t) dt \right)^2 \right)^{1/2} \geq R(z).$$

Write $z_j = e^{i\theta_j} \tanh t_j$, where $t_j = \operatorname{arctanh} |z_j|$ (with an arbitrary phase when $z_j = 0$). The synchronized radial path

$$\gamma_j(t) = e^{i\theta_j} \tanh(tt_j)$$

has constant metric speed $(\sum_j t_j^2)^{1/2}$. Its length is $R(z)$, so the lower bound is attained. $\square$

Direct duality for the displayed metric gives

$$Q_f(w)^2 = \sum_{j=1}^n (1 - |w_j|^2)^2 \left| \frac{\partial f}{\partial w_j}(w) \right|^2. \quad (2)$$

Theorem 2. For every $z \in \mathbb{D}^n$,

$$\omega_0(z) = \omega(z) = \rho(z, 0) = \left(\sum_{j=1}^n \operatorname{arctanh}^2 |z_j| \right)^{1/2}.$$

In particular, the Poincare-distance upper bound in (1) is sharp at every point.

Proof. The result is immediate for $z = 0$, so assume $R = R(z) > 0$. Set

$$t_j = \operatorname{arctanh} |z_j|, \quad a_j = \frac{t_j}{R},$$

and choose $\eta_j \in \mathbb{T}$ so that $\eta_j z_j = |z_j|$ whenever $z_j \neq 0$; the choice is immaterial when $z_j = 0$. For $0 < s \leq 1$, define

$$F_s(w) = \sum_{j=1}^n a_j \operatorname{arctanh}(s \eta_j w_j), \quad w \in \mathbb{D}^n, \quad (3)$$

where $\operatorname{arctanh} \zeta = \frac{1}{2} \log \frac{1+\zeta}{1-\zeta}$ is taken in its analytic disk branch.

Using (2),

$$\begin{aligned} Q_{F_s}(w)^2 &= \sum_{j=1}^n a_j^2 \frac{s^2(1 - |w_j|^2)^2}{|1 - s^2 \eta_j^2 w_j^2|^2} \\ &\leq \sum_{j=1}^n a_j^2 = 1. \end{aligned}$$

Indeed,

$$|1 - s^2 \eta_j^2 w_j^2| \geq 1 - s^2 |w_j|^2 \geq s(1 - |w_j|^2),$$

the last difference being $(1 - s)(1 + s|w_j|^2) \geq 0$. Also $F_s(0) = 0$.

At $s = 1$ we therefore have F_1 in the normalized Bloch unit ball, and

$$F_1(z) = \sum_j a_j \operatorname{arctanh} |z_j| = \frac{1}{R} \sum_j t_j^2 = R.$$

Thus $\omega(z) \geq R$.

If $s < 1$, then F_s extends holomorphically to a neighborhood of the closed polydisk. Its first derivatives are bounded there, so (2) implies $Q_{F_s}(w) \rightarrow 0$ as w approaches the distinguished boundary. Hence $F_s \in \mathcal{B}_{0*}(\mathbb{D}^n)$. Since

$$F_s(z) = \sum_j a_j \operatorname{arctanh}(s|z_j|) \longrightarrow R \quad (s \uparrow 1),$$

we obtain $\omega_0(z) \geq R$. Combining these inequalities with (1) and Lemma 1 proves all equalities. $\square$

4 Operator-theoretic consequences

Let μ be a positive continuous weight, $\psi \in H(\mathbb{D}^n)$, and $\varphi = (\varphi_1, \dots, \varphi_n)$ a holomorphic self-map of $\mathbb{D}^n$. Define

$$\mathcal{R}_\mu(\psi, \varphi) = \sup_{z \in \mathbb{D}^n} \mu(z) |\psi(z)| \left(\sum_{j=1}^n \operatorname{arctanh}^2 |\varphi_j(z)| \right)^{1/2}.$$

Corollary 3. *The weighted composition operator $W_{\psi,\varphi}f = \psi(f \circ \varphi)$ is bounded from $\mathcal{B}(\mathbb{D}^n)$ to $H_\mu^\infty(\mathbb{D}^n)$ if and only if $\psi \in H_\mu^\infty(\mathbb{D}^n)$ and $\mathcal{R}_\mu(\psi, \varphi) < \infty$. In that event,*

$$\|W_{\psi,\varphi}\| = \max\{\|\psi\|_{H_\mu^\infty}, \mathcal{R}_\mu(\psi, \varphi)\}.$$

The identical statement holds with $\mathcal{B}_{0}(\mathbb{D}^n)$ as domain.*

Proof. Substitute Theorem 2 into the two parts of [1, Theorem 3.3], which express the exact norm using ω and ω_0 , respectively. $\square$

The source’s compactness condition on the polydisk uses $\sum_j \operatorname{arctanh} |\varphi_j(z)|$. Since for nonnegative vectors

$$\left(\sum_j t_j^2\right)^{1/2} \leq \sum_j t_j \leq \sqrt{n} \left(\sum_j t_j^2\right)^{1/2},$$

its little-oh condition is equivalently expressed with the exact quantity $R(\varphi(z)) = \omega(\varphi(z))$.

5 Novelty check

The run’s cheap indexes contained no packet for arXiv:2208.02147 or this point-evaluation extremal. Bounded arXiv searches on 9 August 2026 used the exact source question, title, and identifier, together with “Bloch polydisk”, “point evaluation”, “ $\omega(z)$ ”, “Poincare distance”, and “ $\operatorname{arctanh}$ ”. They located the source and older generalized-Bloch composition papers, but no later answer to Open Question 1. This supports possible novelty but is not an exhaustive bibliographic guarantee.

References

- [1] R. F. Allen, *Weighted composition operators from the Bloch space to weighted Banach spaces on bounded homogeneous domains*, arXiv:2208.02147 (2022).